Jia Zeyue

Email | GitHub | GitLab

Education

National University of Singapore *August 2025 – Present*
Faculty of Science | *GPA: 4.64/5.00* *Singapore*
Master | *Data Science and Machine Learning*

University of Science and Technology of China *September 2021 – June 2025*
School of the Gifted Young *Hefei, Anhui, China*
Bachelor | *Mathematics and Applied Mathematics*
Coursework | *Programming Language Design and Program Analysis*

Research Interests: Programming Languages, Formal Methods, Software Engineering, LLM/Agentic Systems for Software Development

Research Experience

Structured Representation and Analysis of Agent Skills

Supervisor: Prof. Chengpeng Wang 2026 – Present

- Developing structured representations of agent skills to expose intermediate steps, inter-step dependencies, and semantic constraints for programmatic analysis
- Implemented execution and profiling mechanics for the structured representation of skill
- Designing runtime scheduling and routing techniques for optimized skill execution

Formal Verification of Distributed Protocols Using Veil

Supervisor: Prof. Ilya Sergey Jan 2026 - June 2026

- Formalized the protocol model from the Autobahn paper in Veil, adapting the protocol's state, transition, and commit behavior into Veil's verification framework
- Constructed an executable model-checking artifact and validated it against a canonical commit trace using Veil's `#model_check`

A Theorem Proving Framework Integrating LLM Agentic Systems and Formalized Intermediate Language

Thesis Sept 2024 – June 2025
Supervisor: Prof. Guangzhong Sun

- Developed an LLM-assisted theorem-proving framework using an intermediate formal language for solving mathematical olympiad problems
- Applied chain-of-thought and few-shot prompting to translate natural-language mathematical problems into formal statements in MantLe
- Built an interaction layer between LLM agents and the MantLe compiler using a Continuation-Passing-Style Agent Monad for communication, action, and feedback
- Earned a [4.0/A] grade in the thesis defense

Neural Symbolic Reasoning in Mathematics

USTC–Microsoft Research Asia Innovation and Practice Project May 2024 – June 2024
Supervisor: Dr. Fan Yang

- Investigated LLM-based auto-formalization and automated proof completion for mathematical reasoning
- Used *ToRA* with few-shot and chain-of-thought prompting for Lean auto-formalization and proof-step generation
- Evaluated the approach on 50 problems sampled from miniF2F, analyzing performance in auto-formalization and proof completion

Complex Knowledge Representation and Reasoning with Applications in IMO Automatic Problem-solving
USTC Knowledge Computing Lab Jan 2023 – Jan 2024
Supervisor: Prof. Yi Zhou
1M CNY Granted from the National Natural Science Foundation of China

- Studied assertional logic for representing declarative and procedural mathematical knowledge and developed methods for knowledge acquisition and reasoning
- Modeled theorems and problem-solving strategies as inference rules and implemented a reasoning system supporting rule- and method-based search

Application of an Assertion Logic-based Generalized Inference Engine in Mathematical Competitions
USTC Undergraduate Research Program July 2023 – Nov 2023
Supervisor: Prof. Yi Zhou

- Implemented a reasoning engine based on a knowledge representation language called MantLe and the methodologies of assertional logic and knowledge modeling
- Developed a structured and highly scalable parser using pushdown automata, employed BFS with weighted sorting for method and rule matching in the engine, and realized post-matching theorem replacement and generation through the JavaScript prototype system
- Achieved 70.8% accuracy on 120-problem dataset

Compiler and Open-Source Experience

MoonBit Language Core Repository Contribution
Open-source Contributor May 2025 – Present

- Contributed three accepted pull requests to the MoonBit core repository, addressing performance and consistency issues
- Designed a new *template* language feature for MoonBit and am implementing compiler support for the feature

GHCup Mirror @ USTC Open-Source Software Mirror Site
Maintainer Dec 2024 – Present

- Maintained GHCup mirror infrastructure, including compiler-channel synchronization and bootstrap support for mirror-dependent environments

Competition and Projects

iGEM Competition
24-member Team, leadership, Silver Medal Apr 2023 – Nov 2023
36-member Team, advisor, Gold Medal Apr 2025 – Nov 2025

- Developed a mathematical model for improving L-arabinose isomerase expression in *Kluyveromyces marxianus* and presented the project at the Grand Jamboree in Paris
- Led website team to build team wiki to present our results

Extracurricular Activities & Awards

TA of Computer Programming Class Sept 2023 – Feb 2024

- Held office hours for 50+ students and implemented an online grading system for assignments and instant feedback
- 2022 The Chinese Mathematics Competitions, First Prize in Category A (Ranking 57/1389, Top 4%)

Skills

Formal Methods: Lean 4, Agda, Veil
Programming Languages: Haskell, OCaml, MoonBit, C/C++, Python, JavaScript